



Structured deep learning based object-specific distance estimation from a monocular image

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Abstract

Distance calculation is a critical link in the research fields of object trajectory prediction, automatic driving obstacle avoidance, and so on. However, the research on distance using deep learning methods has yet to attract wide attention. The accuracy of traditional distance estimation algorithms based on the optical principle and mathematical modeling is low in practical applications, mainly the curve or slope of the road surface. This paper addresses the challenging distance estimation problem by developing an end-to-end structured model to directly predict the distance for objects in a given image. Besides, the traditional mathematical modeling process is replaced by this learning-based method. To facilitate the research on this task, we construct the extended distance datasets by KITTI (Karlsruhe Institute of Technology and Toyota Technological Institute) and NYU (Nathan Silberman, Pushmeet Kohli, Derek Hoiem, Rob Fergus) Depth V2 distance datasets. Experimental results demonstrate that the structured learning model has higher accuracy than the traditional algorithm in different distance ranges and better performance for curves and ramps. Moreover, improving neural network performance will be the direction of improving the model in the future.

Keywords Distance estimation · Convolutional neural network · Monocular image · Structured deep learning · Conditional random field · Machine learning · Computer vision · Single camera

1 Introduction

With the development of autonomous driving technology, perceiving the surrounding environment and analyzing it has gradually become an important research subject. Therefore, computer vision as a crucial part attracted the extensive

attention of researchers. As one of the computer vision technology applications, object-specific distance estimation is an essential part of autonomous driving [1]. By estimating the distance of various targets on the road, such as pedestrians and vehicles, it can provide important reference information for the automatic driving system to avoid collisions, maintain a safe distance between vehicles, speed control, and cruise and other vital functions. Traditional distance prediction measurement could be divided into the initiative and passive range measurements [2, 3]. Initiative ranging can also be called direct ranging. The distance to the specific object is determined by emitting light with a particular texture to illuminate the object, analyzing the texture deformation of the reflected light from the object, or directly measuring the travel time of the light. In general, the result of active ranging is accurate, but the equipment required is expensive and large, and the data of active ranging usually needs further processing to get the corresponding distance information. More importantly, it is necessary to recognize some icons or signs and calculate the corresponding distance in automatic driving. For example, measuring the distance to the zebra crossing is required in some cases, while the radar

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range cannot directly calculate the corresponding distance. Passive ranging methods determine the distance of an object by detecting its natural light radiation and analyzing it, such as stereo vision [4], motion ranging [5], defocus ranging [6, 7], inverse perspective mapping (IPM) and other algorithms. The passive ranging algorithm attempts to establish a mathematical model according to the optical principle to express the relationship between the target distance and information from reflected light.

However, such mathematical models were established in ideal conditions. The inverse perspective algorithm needs to ensure the axis of the camera and the road level and cease to be effective in case of slope or turn. In the range process, stereo vision and motion distance detection methods must find feature points for feature matching. However, the accuracy of the range cannot be guaranteed due to feature point occlusion and matching errors in practical applications. The passive ranging algorithm performs poorly in a complex environment, and its ranging accuracy depends on the camera used. It has a poor ranging performance for objects that are too close or far away, especially targets over 40 m or less than 4 m. The traditional passive ranging algorithm tries to get the target range directly according to a specific optical feature. Still, the accuracy could not be guaranteed in practical application due to many interference factors. Such algorithms ignore the character of depth information contained in RGB (Red, Green, Blue) images in the ranging process. There are many successful applications of neural network methods in visual perception, such as object detection, object classification, object segmentation, etc. However, [8] there are few pieces of research on using deep learning techniques for object-specific distance estimation. The main reasons could be the lack of object distance datasets and a standard definition of target distance.

In this paper, we propose a structured deep-learning model to address the challenging problem that directly calculates the distance of objects in a given RGB image. The structured, end-end model consists of the convolutional neural network (CNN) and continuous condition random field (CCRF). Here, we bridge the gap of using the deep learning method for distance estimation problems by jointly exploring CNN and CCRF, denoting this new model as a structured convolutional neural field (SCNF) model. The passive ranging algorithms mentioned above do not depend on hardware devices, so they can be called soft computing algorithms. SCNF overcomes the shortcomings of the soft computing techniques described above, including the following: SCNF can adapt to complex environments without any assumptions; SCNF computes faster and can be applied in real-time scenarios; SCNF has higher accuracy in different ranging ranges and can also accurately measure objects with a distance of less than 5 m. SCNF does not require expensive hardware equipment and has a broader application prospect

in practical applications. Finally, compared with the active ranging algorithms mentioned above, SCNF has a broader application prospect for the following reasons: SCNF model does not rely on expensive equipment; The SCNF model is an end-to-end model that can get results directly without any processing of the results. Compared with the CNN distance prediction model, the structured model considers the relationship between the target and the background in a given image and can reduce the influence of background pixel blocks on the prediction results, so it has higher prediction accuracy. This will be demonstrated in the ablation experiment section. In previous similar probabilistic regression problems, some used Markov random field (MRF) methods [9, 10], and others used CCRF methods [11]. Compared with MRF, CCRF requires less computation and can be calculated directly without an approximate inference method. Therefore, the CCRF method is adopted in this paper. In this paper, a distance dataset is constructed based on the KITTI and NYU Depth V2 datasets, and experiments prove the effectiveness of the SCNF model.

In summary, the main contributions of object distance estimation are concluded as follows:

- (1) We propose a structured deep learning model with a novel end-to-end framework to directly predict distance from given objects on RGB images without any prior conditions.
- (2) We constructed the CCRF model according to the relationship between adjacent image patches, and the distance calculation formula is theoretically solved.
- (3) We construct the extended KITTI and NYU object-specific distance datasets to facilitate the training and evaluation of this task. The experiment results demonstrate that our proposed method performs better than traditional approaches.

The remaining part of the paper will be discussed in the following sections: related work, ranging algorithm detailed, paper experiment description, and experimental results analysis, at the end of the conclusion.

2 Related work

It is a pioneering work to use deep learning methods for monocular distance measurement. Our model combines the advantages of distance estimation algorithms, target detection, and depth recovery methods. In this section, we briefly review the practices mentioned above.

2.1 Traditional distance estimation algorithm

Traditional distance estimation methods are based on optical principles to establish a mathematical model to describe the geometric relationship between the image and the actual object. The traditional distance measurement model using 2D images to calculate 3d distance is an ill-conditioned mathematical model, which requires strict prior conditions to calculate target distance. This kind of model has poor measurement accuracy and limited application scope. One of the traditional ways to estimate distance for a given object was the inverse perspective mapping (IPM) algorithm [12, 13]. Due to the use of the geometric similarity principle, it is necessary to ensure the camera axis and road level during the measurement process. IPM would fail in cases where objects are on a sloping or curved road. The depth from defocus (DFD) method uses two or more images with different degrees of dispersion to get the distance information of the object by comparing the blur degree of the images. It is unsuitable for practical use because of its strict prior conditions and low-ranging accuracy. All the methods based on the optical range must obtain the camera parameters in advance. The camera lens dramatically affects the range effect, so the results obtained using the same algorithm for different cameras differ significantly. In practice, this kind of algorithm would fail to predict the distance of objects out of distance bounds, too far or too close, due to the lens. Because of various problems in the practical application of the optical model, researchers began to use machine learning methods to improve the visual ranging model. For example, Ref. [14] support vector machine (SVM) was trained to predict the distance at the size of the bounding box of an object. Other researchers attempt to establish a neural network for distance estimation DisNet [15]. try to train a CNN-based model (YOLO) instead of a conventional optical method for distance estimation. The above process attempts to improve the traditional way by using machine learning to find the geometric relationship between the target image and the geometric distance [8]. Zhu used neural networks for distance estimation for the first time. The neural network model proposed by ZHU is an end-to-end ranging model, and the distance can be obtained directly without setting any prior conditions. The based model utilizes a CNN model to regress the object distance, and the enhanced model tries to add geometric information to improve the performance. This model proves that the neural network can be applied to the target range.

Other distance estimation algorithms use multiple images of the same object to analyze the distance to the target. Motion parallax algorithms acquire a series of object images at different times and spatial locations based on an imaging sensor. The distance surface information of the object is extracted according to the time and space changes of the

two-dimensional image sequence of the object. Another similar distance estimation method is the stereo vision algorithm. The basic principle of the stereo vision distance estimation algorithm is to observe the same scene from two (or more) viewpoints to obtain perceptual images from different perspectives and to obtain the scene by calculating the positional deviation (that is, parallax) between image pixels through the principle of triangulation. This type of method needs to find the corresponding features of the image and calculate the displacement between them. Extracting features of objects is a very time-consuming and complex task. These two ranging methods are limited by the occlusion problem (that is, the object is visible on one scene but not on the other scene, and the corresponding features cannot be found, so the depth of the object cannot be calculated). Meanwhile, their real-time ranging accuracy is low.

2.2 Other visual perception algorithms

Computer vision algorithms inspire the SCNF proposed in this paper in other fields. At present, the commonly used target detection models are SSD [16] and YOLO [17], the neural network model for target detection and target classification mentioned in [18–20], and the image segmentation algorithm R-CNN. Although these algorithms are not used for target distance calculation, these models have a certain similarity; they use the neural network to extract image features and perform calculations with the extracted image features.

The problem of depth recovery is another area related to distance estimation. Many researchers have tried to solve the depth recovery problem using neural network models [21–25], computing the corresponding depth map from a given RGB image. On the problem of depth recovery using a monocular camera, supervised learning methods are often limited by the difficulty of obtaining real depth and insufficient training samples. In contrast, unsupervised learning methods are challenging to guarantee computational accuracy. In the problem of depth estimation, due to the need to calculate the depth of each pixel, the model needs to process a large amount of data, and the calculation amount is enormous, which cannot be applied to real-time target distance estimation, especially in the field of autonomous driving.

Farabet et al. [26] proposed a multiscale CNN for scene labeling, where a conditional random field (CRF) was used for local optimization, and Tompson et al. [27] proposed a CNN and a Markov Random Field (MRF) The combined model is used for human gesture recognition. Recently, some researchers have begun using CRF to solve the deep recovery problem. Liu [28] proposed to combine CRF and CNN to obtain a closed-form solution for calculating pixel points, which accelerated the calculation speed and training speed. These image-processing algorithms inspired us

to address problems of object distance estimation by deep learning methods.

In addition to the above algorithms, some excellent algorithms in recent years have provided us with a broader idea. In the paper, Refs. [29, 30], based on the original deep learning model, the model performance is improved by combining other algorithms. It is a common technology in current computer vision to use deep learning to extract image features and then further process image features [31, 32]. This paper adopts a similar image-processing idea. Similarly, the methods of combining traditional mathematical models with deep learning to generate new models [33, 34] also inspire us well. The technique of image content analysis [35, 36] combined with historical data has gradually become a hot topic, which will be the direction of model improvement in the future.

3 Our approach

SCNF proposed in this paper is a Boosting model, including the CNN and CCRF models (Fig. 1). Among them, the CNN part extracts images' depth features and calculates the distance's regression value. The output value of the CNN part is used as the input parameter of the CCRF part to calculate the final target distance. Before the distance estimation process, an object detection algorithm (such as YOLO) must extract the specific object image boundary from the given image. In order to weaken the influence of the background, the object image needs to be evenly divided into four patches for distance estimation. We define the distance of each image patch as the weighted average of the depth values of all pixels within the pixel patch. Define the object's distance as the average of the distances of the four image patches. Endow pixels closer to the center of the integrated object image more immense weight. Actually, each patch of the object image is the input of the integral model.

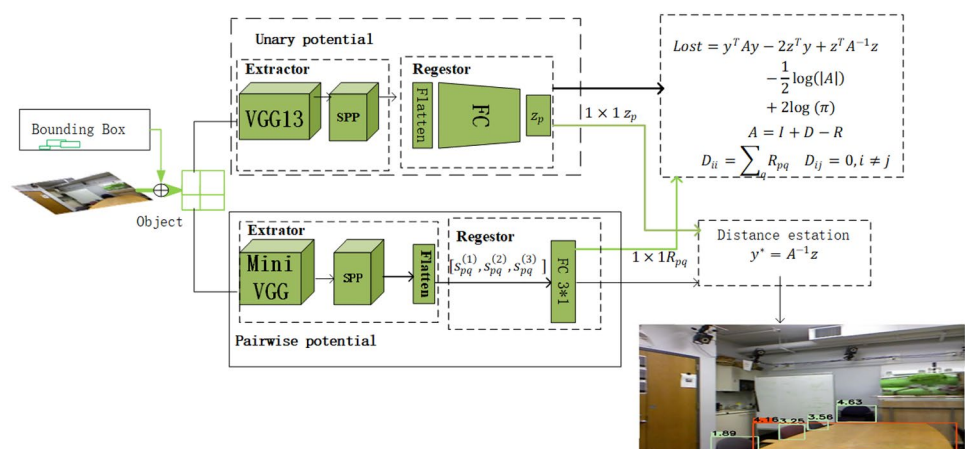
3.1 CNN model

The CNN part of SCNF has two functions: calculate the regression value of each image patch and extract the eigenvalue of each patch of the object image. The CNN model in SCNF is divided into unary and pairwise parts to implement corresponding functions. The unary part calculates the regression value and is used to fit each image patch distance value. The pairwise part extracts the feature vector of the target image block and calculates the similarity of each part of the image. This paper proves the effectiveness of each part of the neural network model through ablation experiments in 4.2. Since we divide the image into four image patches at a time, the object distance can be accurately calculated from the closed-form solution calculated by CCRF without an estimation method. Although the size of the object image is unfixed, in order to reduce the influence of interpolation, we do not resize the object image but use the SPP method to generate a fixed-size feature vector [37].

3.1.1 Unary part

The neural network model of the unary part includes two parts: image feature extraction and regression calculation. Among them, the feature extraction part is a VGG-like network model consisting of 10 convolutional layers for extracting features. There is a maximum pooling layer between every two convolutional layers. Except for the last convolutional layer, all hidden layers use ReLu as an activation function. The final convolutional layer uses the SPP pooling layer to connect the particular regression value calculation part. The regression calculation part uses five fully connected (FC) layers (with layers of size {2048, 1024, 512, 512, 1}) to calculate the regression value corresponding to the feature vector. Except for the last fully connected layer, each layer uses ReLu as the activation function and uses Dropout to prevent overfitting.

Fig. 1 Structured model



The problem studied in this paper is unique. The input to the model is a partial image extracted from the original image. Using the interpolation method to change the image size in the image preprocessing stage will bring much noise, especially for distant targets; the image size is petite, and if the image size is adjusted by the interpolation method, it will cause significant interference with the prediction results. The image size is not changed in the preprocessing stage, so it is impossible to continue increasing the neural network's depth. Since the input image size is not fixed, the extracted feature vector is not fixed, and SPP pooling is used to obtain a fixed-size feature vector. The SPP model proposed by He. [37] needs to evenly divide the image into 1, 4, and 16 blocks for maximum pooling. The object processed in this paper is a quarter of the original target image, so the idea is only segmented and pooled twice, resulting in a feature vector of size $512 \times 1 \times 5$.

3.1.2 Pairwise part

Since all image blocks belong to the same target, the eigenvalues of each image block should be as close as possible. Therefore, we need to calculate the difference in feature values of different image blocks in the pairwise part. Some depth restoration algorithms [16, 28] use color difference, color histogram difference, and texture difference as image patch features. In contrast, this paper uses the feature vector proposed by the CNN model in the pairwise part to represent each image patch. The feature extractor can be dynamically adjusted to better express the image features by the neural network method instead of some specific indicators.

The neural network model of the pairwise part has a similar structure to the neural network of the unary function and is also composed of two parts: feature extraction and regression calculation. The feature extraction model consists of six convolutional layers, each followed by a max-pooling layer except the last pooling layer. The input image size is not adjusted for the same reason as in the unary part, and the SPP pooling method is used to obtain four feature vectors of size 1×3 corresponding to four image blocks.

After obtaining the corresponding feature vectors of each image block, we need to calculate the difference between the feature vectors further. One of our optimization objectives is minimizing the eigenvalue difference between different image blocks. First, we construct a $4 \times 4 \times 3$ difference value matrix S . Second; we use a fully connected layer of size 3×1 to normalize S to get the final difference matrix R , whose size is 4×4 . Specifically, select Softmax as the activation function for the fully connected layer. In order to prevent the weight of the fully connected layer from tending to 0, we set the minimum value of the weight of the fully connected layer to 0.1.

$$S_{pq}^k = \frac{2}{1 + e^{-\|s_p^{(k)} - s_q^{(k)}\|_2}} - 1 \quad k = 1, 2, 3. \quad (1)$$

In Eq. (1), $s_p^{(k)} s_q^{(k)}$ represent the k_{th} feature value of image block p and q in the image respectively, $\|\cdot\|_2$ is a Two-norm formula.

3.2 CCRF model

The CCRF model of SCRF is modeled according to the relationship of adjacent image patches. We prove the effectiveness of CCRF by experiment.

3.2.1 Specific-object distance

How to define the target distance reasonably is the key to the research problem, and there are many methods for specifying the distance worth referring to. In the depth recovery task of [10, 28], the image is regarded as a combination of $N \times N$ superpixels, and the depth of the superpixel center pixel is defined as the depth of the superpixel; in [8], the target distance is defined as a random selection in the target image. The depth of a point is the target distance. In this paper, referring to the previous definition of target distance, the weighted average of all pixel depth values in the target image is used as the target distance. First, the target image is average divided into four blocks from the center, and each image block is regarded as a large superpixel block; A weighted average is performed to obtain the target distance of the corresponding image block. Finally, the average of the distances of the four image patches is calculated as the overall target object distance.

The bounding box given by the target detection algorithm includes a specific range of background images. Generally, the part between the bounding box provided by the target detection algorithm and the object margin is regarded as the background image. For a target image, points further away from the center of the image are more likely to be background pixels. Therefore, the weight of each pixel should be inversely proportional to the distance from the center of the image. Compared with the Chebyshev distance and Euclidean distance, the Manhattan distance can better describe the descending relationship of weight and is easy to calculate and realize. Therefore, this paper chooses the Manhattan distance to calculate pixel weight. This part of the image is distributed around the bounding box so that we can set the weight according to the distance between the pixel and the center of the image, and the pixels farther from the center are given smaller weights. In this paper, the distance refers to the Manhattan distance. The specific definition method of the weight matrix is as follows. First, set the weight corresponding to the center pixel of the image as 0.5. Set the furthest distance center pixel in the image of the Manhattan distance

to c ; If the number of pixels with Manhattan distance i is N_i , then the corresponding weight of the pixels with Manhattan distance i is:

$$W = 0.5 \times \frac{c-i+1}{\sum_{j=1}^c j} \times \frac{1}{N_i} \quad (2)$$

$$W_{center} = 0.5$$

The depth matrix W corresponding to the pixel block is the distance corresponding to the target image block.

3.2.2 Distance estimation

Define $x = [x_1, x_2, x_3, x_4]$ for a given target image, the corresponding image block which is suitable for target image x ; $y = [y_1, y_2, y_3, y_4] \in R^4$ represents the corresponding distance values of the four image blocks in image x . According to the definition of CCRF, the relationship between image x and distance y can be expressed as a posterior probability formula:

$$P(y|x) = \frac{1}{Z(x)} e^{(-E(y,x))} \quad (3)$$

where $E(y, x)$ is the energy function and $Z(x)$ is the normalized factor. Since the distance y is a continuous variable, Z is defined by integration as:

$$Z(x) = \int_y \exp(-E(y, x)) dy \quad (4)$$

The energy function of CCRF is defined as the combination of the unary potentials U and the pairwise potentials V over the image patches:

$$E(y, x) = \sum_{p \in N} U(y_p, x) + \sum_{(p,q) \in S} V(y_p, y_q, x) \quad (5)$$

The unary term U aims to regress the distance of a single image patch. The pairwise term V encourages image patches of one object to take similar feature vectors. In Eq. (5), N is the number of segmentations of the target image. In [10], in order to achieve depth recovery, the image is divided into N superpixel blocks. The depth calculation involves the inverse calculation of $N \times N$ dimensional state transition matrix, and the calculation time complexity is $O(N^3)$. This means deep recovery tasks using CCRF must adopt various estimation methods, such as mean-field theory [28]. However, the distance estimation task only divides the image into four parts. That is, N is 4, which can be calculated accurately without any valuation method. We describe the details of potential functions involved in the energy function in Eq. (5) below.

- (1) The unary potential is constructed from the output of a CNN by considering the least square loss in Eq. (6).

$$U(y_p, -z_p; \theta) = (y_p - z_p(\theta))^2, p = 1 \dots 4 \quad (6)$$

Here $z_p(\theta)$ represents the regressed distance of image patch x_p by the unary part of CNN parameter θ . y_p is the actual distance of image patch x_p . One of our optimization goals is to minimize the difference between the regression value and the actual distance.

- (2) The pairwise potential is constructed in Eq. (7) to describe the difference between adjacent image patches.

$$V(y_p, y_q, x; \beta) = \frac{1}{2} R_{pq} (y_p - y_q)^2, p, q = 1 \dots 4 \quad (7)$$

Here R_{pq} is the output of the CNN in the pairwise part from the adjacent image patch (x_p, x_q) . y_p, y_q is the actual distance. β is the parameters of the pairwise part we need to optimize in CNN.

Then the complete energy formula can be obtained.

$$E(x, y) = \sum_{p \in N} (y_p - z_p)^2 + \sum_{(p,q) \in S} \frac{1}{2} R_{pq} (y_p - y_q)^2 \quad (8)$$

For the convenience of calculation, Eq. (8) is re-expressed in matrix form:

$$A = I + D - R \quad (9)$$

I is an $N \times N$ identity matrix, and R is a state transition matrix between blocks. D represents a diagonal matrix, where:

$$D_{pp} = \sum_q R_{pq} \quad (10)$$

A is a symmetric positive definite Laplace matrix. The matrix expression of our energy formula is:

$$E(x, y) = y^T A y - 2z^T y + z^T z \quad (11)$$

In this paper, all N in the above formula is 4. After completing the definition of the energy function, the normalization factor can be obtained:

$$\begin{aligned} Z(x) &= \int_y e^{-E(x,y)} dy \\ &= e^{-z^T z} \int_y e^{-y^T A y + 2z^T y} dy \\ &= \frac{\pi^2}{|A|^{\frac{1}{2}}} e^{z^T A^{-1} z - z^T z} \end{aligned} \quad (12)$$

The posterior probability formula can be expressed as:

$$P(y|x) = \frac{|A|^{\frac{1}{2}}}{\pi^2} e^{-y^T A y + 2z^T y - z^T A^{-1} z} \quad (13)$$

According to the principle of maximizing the posterior probability, we get the lost function to be optimized in the end:

$$\begin{aligned} \text{loss} &= -\log P(y|x; \theta, \beta) \\ &= y^T A y - 2z^T y + z^T A^{-1} z - \frac{1}{2} \log(|A|) + 2\log(\pi) \end{aligned} \quad (14)$$

Among them, θ and β are the learned parameters of the unary and pairwise parts of the neural network model, respectively. The calculation of the target distance is to find the y corresponding to the maximization of the posterior probability $P(y|x)$. Since $-\log P(y|x; \theta, \gamma)$ is a convex function, we can solve it by differential methods.

$$\begin{aligned} y^* &= \operatorname{argmin}_y \{-\log P(y|x; \theta, \gamma)\} \\ \frac{\partial y^T A y - 2z^T y}{\partial y} &= 0 \Rightarrow y^* = A^{-1} z \end{aligned} \quad (15)$$

Finally, y^* is the predicted value for a patch of the given image.

4 Evaluation

Training and validation models must be trained and validated on indoor and outdoor data sets. The difference between the two data sets lies in the illumination and distance range. Indoor data sets include NYU Depth V2, Scene Flow Datasets, and SUN-RGB-D (A RGB-D Scene Understanding Benchmark Suite). Where SUN-RGB-D is the simulation data set, Scene Flow Datasets are rendered by various composite sequences. NYU has processed RGB images with labels and depth of completion. Since the simulated image cannot completely simulate the illumination conditions of the actual images, the NYU data set is chosen here. Commonly used outdoor data sets are KITTI and nuScenes. Since the experiments in related work [25, 28] were trained and tested on the KITTI data set, KITTI was used as the outdoor data set in order to verify experimental results better. We constructed our distance data set based on the currently available public data sets KITTI [38] and NYU Depth V2 [39] for training and testing. The Adam+SGD optimization strategy was applied to the dataset constructed by NYU and KITTI, respectively. Adam was first used for optimization; the initial learning rate was set to 0.001 and exponentially decayed after ten epochs. Adam optimizer iterated 200 epochs, then SGD was used to continue the training of 100 epochs. After training, SCNF can be used to predict the object-specific distances given any RGB images directly.

The effectiveness of SCNF is proved by quantitative comparison experiments on the data set collected in different scenarios and the traditional ranging algorithm. Compared with SCRF models without CCRF, SCRF models have higher

accuracy and better generality by comparing the ranging accuracy over different range ranges. Finally, the ablation experiment proves the effectiveness of the neural network model. Experimental results show that the proposed model has high computational accuracy and better versatility.

4.1 Data construction

The NYU Depth V2 dataset is a series of indoor Depth maps taken by Microsoft using Depth cameras at target distances of up to 10 ms. KITTI data set is a series of outdoor radar point cloud images. This paper converts point cloud images into depth maps by referring to [38] and the toolkit provided by KITTI's official website. This paper uses the target detection algorithm to process the depth data set, recording the target position and size information detected in each image. The weighted average algorithm in 3.2 generates the data set for distance estimation. The data distribution used in the experiment is shown in Fig. 2.

4.2 Quantitative analysis

After the data set construction is completed in 4.1, it is divided into the training set and test set according to 8:2. The distance prediction model in this paper belongs to the regression model and cannot be evaluated by accuracy and recall rate, so it is necessary to adopt threshold related test methods. Moreover, the metrics used in this paper are the most commonly used method and can be easily compared with the results of others. Referring to the model evaluation method in [21], we adopt the following indicators for quantitative verification:

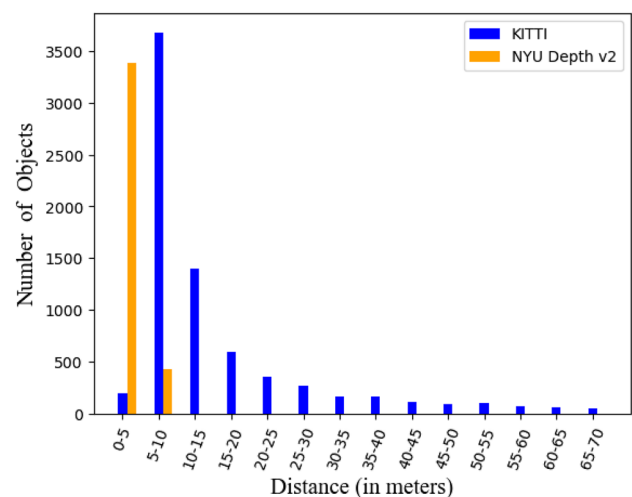


Fig. 2 Distribution of generated object-specific distances

- (1) Absolute relative difference(Abs Rel):

$$AbsRel = \frac{1}{N} \sum_{i=1}^N \frac{|d[i] - d_{pre}[i]|}{d_{pre}[i]} \quad (16)$$

- (2) Squared Relative Difference(Squa Rel):

$$SquaRel = \frac{1}{N} \sum_{i=1}^N \frac{\|d[i] - d_{pre}[i]\|^2}{d_{pre}[i]} \quad (17)$$

- (3) Root of Mean Squared Errors(RMSE):

$$RSME = \sqrt{\frac{1}{N} \sum_{i=1}^N \|d[i] - d_{pre}[i]\|^2} \quad (18)$$

- (4) The root of Mean Squared Errors computed from the log of the predicted distance and the log ground truth distance($RMSE_{log}$):

$$RMSE_{log} = \sqrt{\frac{1}{N} \sum_{i=1}^N \|\log d[i] - \log d_{pre}[i]\|^2} \quad (19)$$

- (5) Accuracy at a certain threshold:

$$Accuracy : \% \text{ of } \max\left(\frac{d}{d_{pre}}, \frac{d_{pre}}{d}\right) = \delta < threshold \quad (20)$$

IPM is one of the classic algorithms in automatic driving. IPM calculates the conversion matrix between RGB image and aerial view using camera parameters. After the object's center point is projected to the coordinates of an aerial view, the forward direction value is used as the estimated distance. The SVR model is a distance measurement method combining machine learning and traditional techniques. Extract the unary part of SCNF as a single CNN model for comparison, and call this model only CNN. These three models belong

to the traditional optical algorithm, the traditional algorithm improved by machine learning, and the neural network model. These three models are the mainstream methods in the current research, and the three benchmark models are the classical algorithms in these three directions. In addition, the three benchmark models can directly use the bounding box of the target object as input data, which is the same as the SCNF model, and the comparison results are representative. The comparison results between the benchmark model and SCNF are shown in Table 1.

By comparing quantitative indicators in Table 1, the SCNF model proposed in this paper has achieved the lowest relative error and the highest threshold accuracy compared with other models. SCNF has good generalization ability and robustness. Both NYU and KITTI data sets perform well, with higher precision than other models. The experimental results show that the structured model built on the original CNN model significantly improved performance. The SCNF model has higher threshold accuracy and lower error in different data sets than the only CNN model. The NYU and KITTI data sets are indoor and outdoor, respectively. The illumination conditions of the two data sets are different, and the target distance of all NYU data sets is less than 20 m. By comparing the accuracy of the SCNF model on the NYU data set and the KITTI data set, it can be found that the accuracy of the NYU data set is higher than the accuracy of the KITTI data set. Moreover, it is difficult to meet its prior requirements in practice for the traditional optical distance estimation model, such as ranging on curved roads. As shown in Fig. 3, while ranging on the curved road surface, IPM and SVR fail to predict the distance. As a comparison, the SCNF model can adapt well to long distances and curved roads.

In addition, this paper analyzes the relationship between actual distance and ranging accuracy. As shown in Fig. 4, the abscissa is the exact distance distribution, and the ordinate is the corresponding prediction accuracy. As the actual distance increases, the accuracy of distance estimation decreases. Among them, the prediction accuracy is above

Table 1 The comparisons of object-specific distance estimation with alternative approaches on the subset of our constructed KITTI-object-detection-based dataset NYU-object-detection-based dataset

Dataset	Method	Higher is better			Lower is better			
		$\delta < 1.25$	$\delta < 1.25^2$	$\delta < 1.25^3$	Abs Rel	Squa Rel	RMSE	$RMSE_{log}$
NYU	SCNF	0.865	0.927	0.952	0.157	0.412	1.155	0.263
	Only CNN	0.808	0.916	0.945	0.187	0.446	1.240	0.285
KITTI	SCNF	0.685	0.911	0.964	0.211	1.271	18.394	0.358
	Only CNN	0.645	0.873	0.947	0.246	1.700	21.242	0.384
Average	SVR	0.379	0.566	0.676	1.472	90.143	24.249	1.472
	IPM	0.603	0.837	0.935	0.390	274.785	78.870	0.403
	SCNF	0.775	0.919	0.958	0.184	0.842	9.773	0.311

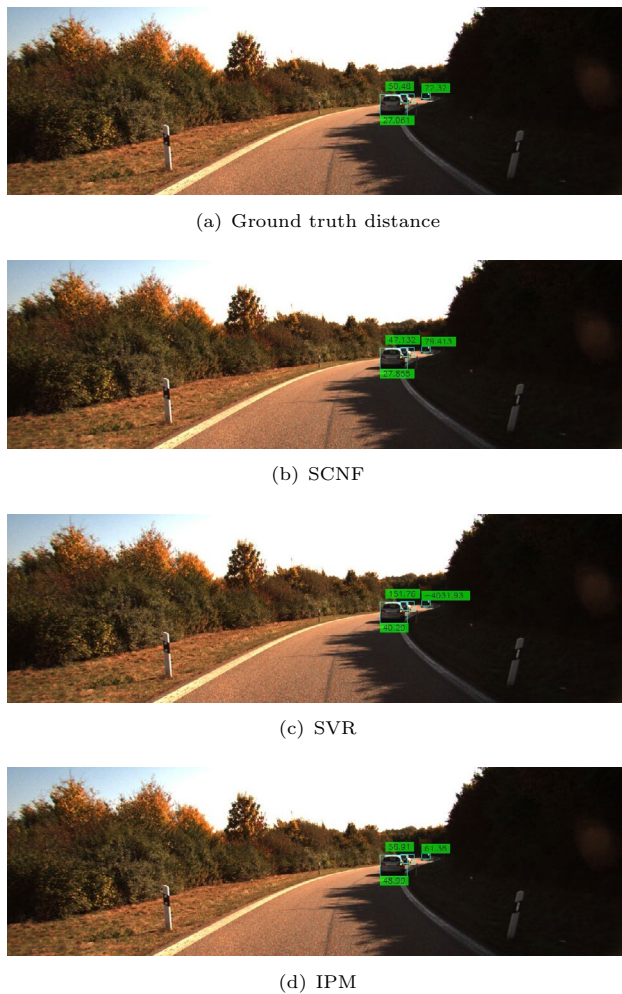


Fig. 3 Examples of comparison of different distance estimation models in curves. We compare the ground truth distance in the top image and the predicted distance in meters using our structured model, SVR, and IPM from top to bottom

0.9 in the 5–40 ms range and above 0.8 in the 0–55 ms range. Among them, the distance with the highest prediction accuracy is 5–20 ms, and the farther the target is, the lower the prediction accuracy is. Because the farther the distance, the fewer pixels of the target image, the less information can be provided, so the farther the distance ranging is more complicated. At the same time, it can be seen from the data set distribution in Fig. 2 that there are relatively fewer training samples with long distances, leading to insufficient training of the model for long-distance targets, which is also one of

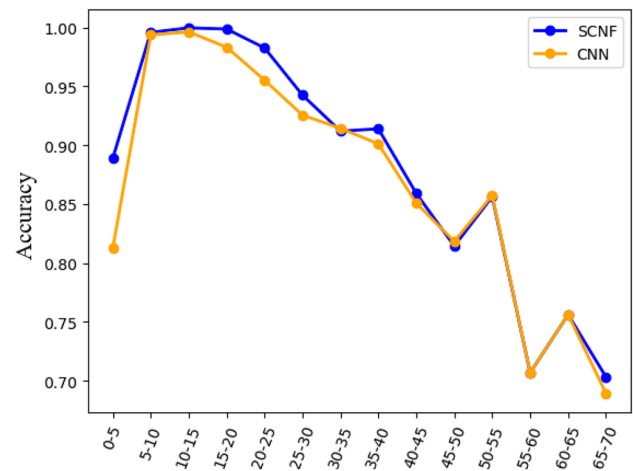
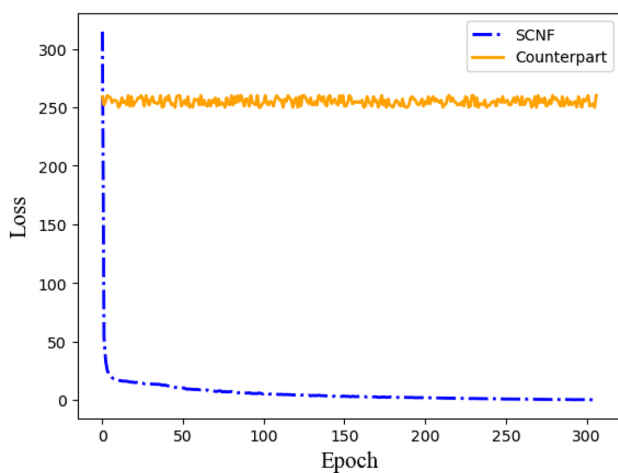
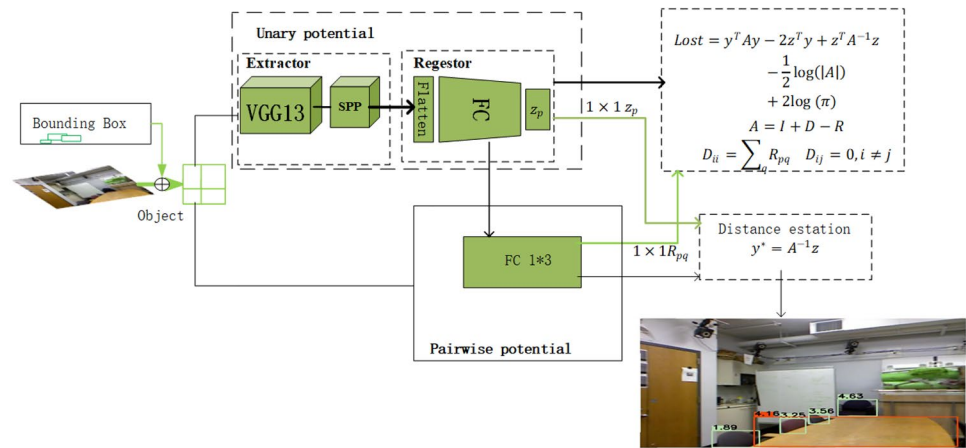


Fig. 4 Accuracy with $\delta < 1.25^3$ on objects with different distances in the KITTI-based dataset (higher is better). The model used for comparison is the CNN part of SCNF

the reasons for the low ranging accuracy of long-distance targets. The bounding box for long-distance targets will be small, limiting the capacity of CNN and making it impossible to use neural networks with larger capacities. So the capacity of the CNN model is the main reason for restricting model performance. Meanwhile, the test results in Fig. 4 proved the effectiveness of CCRF in SCNF.

4.3 Ablation studies

In the neural network model proposed in this paper, a feature extractor is specially designed for the pairwise potential part. In order to verify the effectiveness and necessity of the pairwise part feature extractor, a comparison model is constructed for testing. As shown in Fig. 5, the other parts of the comparison model are the same as the SCNF model. In the pairwise part, features extracted from the unary part are used instead of a separate feature extractor. We tried to train the comparison model on the NYU data set. As shown in Fig. 6, the lost curve of the comparison model oscillates all the time and fails to converge, while the lost curve of the complete SCNF model converges very fast. Experiments show that in model learning, if the pairwise part and the unary part share a feature extraction for learning two different functions will

Fig. 5 Counterpart model**Fig. 6** Result of ablation experiment

lead to mutual interference, resulting in learning difficulties and model convergence problems.

5 Conclusion

This paper studies the deep learning method for object-specific distance detection. There was little research on applying deep learning in the direction of distance estimation before. This paper proposes a structured model called SCNF, which combines CNN and CCRF, to directly predict the distance of objects from RGB images. We trained and tested the SCNF model on constructed dataset extended from KITTI and NYU-depth (V2). This paper selects three representative ranging models: IPM, SVR, and CNN as the benchmark model. Experiments demonstrated that SCNF is superior to the other method. Experiments show that SCNF is very robust, and the accuracy is also higher than that of the benchmark model in cases such as bends and ramps. Ablation experiments proved that it is

necessary to set a separate CNN feature extractor for the pairwise part in SCNF. Due to the small size of the distant target object bounding box, SCNF does not adopt the large-capacity CNN model, which limits the performance of the model. At the same time, the training amount of the long-distance target needs to be bigger, and the geometric information of the image is not combined with other factors limiting the model's performance. In future studies, further improvement of model performance should be considered from the following aspects: modifying the structure of the CNN model, using image segmentation instead of object detection to process the input image, and distance calculation combined with other geometric information.

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